24-S1-Q3

Q: (a) FLD

(b) SVM

Solution (a) ① within-class scatter

$$S_w = S_1 + S_2 = \begin{bmatrix} 223.107 & -13.506 \\ -13.506 & 177.306 \end{bmatrix}$$

$$m_1 - m_2 = \begin{bmatrix} 2.444 \\ -2.595 \end{bmatrix}$$

$$m_1 + m_2 = \begin{bmatrix} -0.718 \\ -0.593 \end{bmatrix}$$

$$w = S_w^{-1}(m_1 - m_2) = \begin{bmatrix} 0.01012 \\ -0.01387 \end{bmatrix}$$

$$w_0 = -\frac{w^T(m_1 + m_2)}{2} = -0.0004797$$

So $g(x) = w^T x + w_0$

$$= \begin{bmatrix} 0.01012 & -0.01387 \end{bmatrix} x - 0.0004797$$

② for $m_1 = \begin{bmatrix} 0.863 \\ -1.594 \end{bmatrix}$

$$g(m_1) = 0.02604534 > 0$$

③ If $g(x) > 0$, classifying x to class 1

If $g(x) < 0$, classifying x to class 2

If $g(x) = 0$, x is on the decision boundary

④ $x_2 = \begin{bmatrix} -0.176 \\ 0.224 \end{bmatrix}$

$$g(x_2) = \begin{bmatrix} 0.01012 & -0.01387 \end{bmatrix} \begin{bmatrix} -0.176 \\ 0.224 \end{bmatrix} - 0.0004297$$

$$= -0.0053677 < 0$$

assign x_2 to class 2

(b) ① SVM w^* calculate

$$w^* = \sum_{i=1} \alpha(i)\, d(i)\, x(i)$$

$$= 0.3784 \times 1 \times [0.287 \quad 0.0421 \quad 0.821]^T$$
$$+ 0.051 \times 1 \times [4.32 \quad -0.572 \quad 3.45]^T$$
$$+ 0.0505 \times (-1) \times [3.922 \quad -1.421 \quad -2.721]^T$$
$$+ 0.3789 \times (-1) \times [0.324 \quad -2.234 \quad 1.122]^T$$

$$= [0.008096 \quad 0.9050 \quad 0.1989]^T$$

② $$w_0^* = \frac{1}{N_s} \sum_{\alpha(i) \neq 0} \left[d(i) - (w^*)^T x(i) \right]$$

$$= \frac{1}{4} \times \left[\left(1 - [0.008096 \quad 0.9050 \quad 0.1989] \begin{bmatrix} 0.287 \\ 0.0421 \\ 0.821 \end{bmatrix} \right) \right.$$

$$+ \left(1 - [0.008096 \quad 0.9050 \quad 0.1989] \times [4.32 \quad -0.572 \quad 3.45]^T \right)$$
$$+ \left(-1 - [0.008096 \quad 0.9050 \quad 0.1989] \times [3.922 \quad -1.421 \quad -2.721]^T \right)$$
$$\left. + \left(-1 - [0.008096 \quad 0.9050 \quad 0.1989] \times [0.324 \quad -2.234 \quad 1.122]^T \right) \right]$$

$$= \frac{1}{4} \times \left[(1 - 0.2037) + (1 - 0.2035) + (-1 + 1.7955) + (-1 + 1.7960) \right]$$

$= 0.7961$

③ classifier

$y = [0.008096 \quad 0.9050 \quad 0.1989] x + 0.7961$

④ test sample $x_3 = [0.225 \quad 2.252 \quad -2.613]^T$

$y = 2.2920 > 0$

So we classify x_3 into class 1 (label 1)